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Review

Pro-environmental behaviors through the lens of the theory of planned behavior: A scoping review

Alexander Yuriev[a,*], Mehdi Dahmen[a], Pascal Paillé[c], Olivier Boiral[a], Laurence Guillaumie[b]

[a] Université Laval, Faculty of Business Administration – Pavillon Palasis-Prince, 2325 Rue de la Terrasse, Quebec City, QC, G1V 0A6, Canada
[b] Université Laval, Centre de Recherche CHU de Québec – Pavillon Ferdinand-Vandry, 1050 Avenue de la Médecine, Local 3645, Université Laval, Québec, Québec, G1V 0A6, Canada
[c] Neoma Business School, 1 rue du Maréchal Juin, 76130 Mont-Saint-Aignan, France

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ABSTRACT

The theory of planned behavior (TPB) allows researchers to identify the determinants of environmental behavior and subsequently target these factors in interventions. Multiple studies on conservation behaviors have recently applied this theoretical framework in both organizational and domestic settings. To shed more light on how the TPB was used in these studies, we conducted a literature review with the following objectives: 1) explore which individual green behaviors were studied though the lens of the TPB, 2) understand how scholars have used the theory and what variance the theory has helped to explain, and 3) formulate recommendations, if necessary, for improving the use of the theory. The review of the results from 126 publications demonstrated that the majority of scholars tend to overlook the importance of identifying and evaluating indirect variables (beliefs) that affect behaviors. More than half of the analyzed articles did not report the amount of explained variance, which undermines the principal strength of the theory. Scholars could obtain more substantial and consistent results if the guidelines regarding the application of the theory are consistently respected. More specifically, four aspects should be considered in the application of the theory: choice of framework, decision to extend the original model, methodology, and results. To help scholars overcome these commonly encountered problems, this article suggests a roadmap with several guiding questions and possible answers.

1. Introduction

The theory of planned behavior (TPB) remains one of the most widely used frameworks for studying individual behaviors. A search using Google Scholar revealed that this theory has been cited almost 90,000 times as of 2019. The TPB (Fig. 1) argues that behaviors stem from individual *intention* and *perceived behavioral control* (PBC). *Intention*, defined as "indicators of how hard people are willing to try… to perform the behavior" (Ajzen, 1991, p. 181), in turn, depends on three direct predictors: attitude, subjective norm, and PBC. *Attitude* is defined as an individual's favorable or unfavorable evaluation of the behavior; *subjective norm* refers to the perceived social pressure towards the behavior; and PBC is the personal assessment of the feasibility of executing the behavior in a given context (Ajzen, 1991). The TPB suggests that the three determinants of intention are influenced by *behavioral*, *normative*, and *control* beliefs, commonly called indirect predictors. *Behavioral* beliefs refer to the perceived advantages and disadvantages of performing a certain behavior; *normative* beliefs are "a person's subjective probability that a particular normative referent wants the person to perform a given behavior" (Ajzen, 2012, p. 441); and *control* beliefs are related to various factors (time, cost, available infrastructures, etc.) that impede or facilitate a behavior.

Since its development almost three decades ago, the TPB has frequently been applied to understand the factors underlying various pro-environmental behaviors: the use of alternative transportation (Muñoz et al., 2016), waste recycling (Echegaray and Hansstein, 2017), saving water (Lam, 2006), energy conservation (Allen and Marquart-Pyatt, 2018), low carbon consumption (Jiang et al., 2019), and others (see Gkargkavouzi et al., 2019 for examples of other domains). Recently, an increasing number of articles have used the TPB to explore the pro-environmental behaviors of employees (e.g., Blok et al., 2015; Boiral et al., 2015; Greaves et al., 2013; Yuriev et al., 2020).

However, there are serious concerns regarding the completeness and efficiency of the TPB for predicting green behaviors. For instance, a

* Corresponding author.
E-mail addresses: Alexander.Yuriev.1@ulaval.ca (A. Yuriev), Mehdi.Dahmen.1@ulaval.ca (M. Dahmen), Pascal.Paille@neoma-bs.fr (P. Paillé), Olivier.Boiral@mng.ulaval.ca (O. Boiral), Laurence.Guillaumie@fsi.ulaval.ca (L. Guillaumie).

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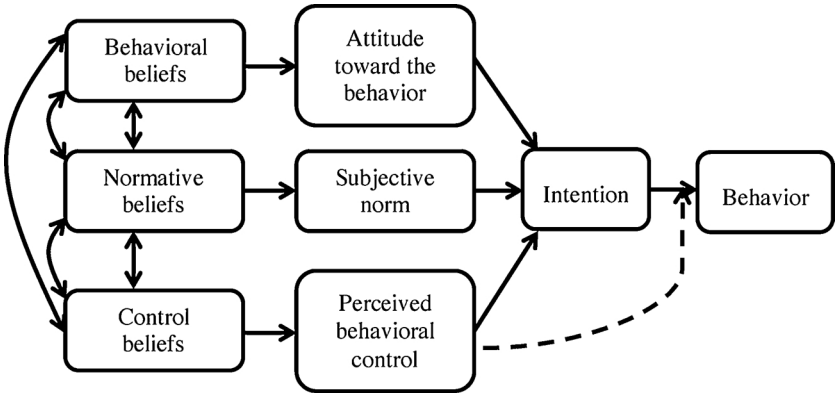


Fig. 1. Theory of Planned Behavior.
Source: Ajzen's official website: http://people.umass.edu/aizen/tpb.diag.html (accessed on the 30th of October 2019).

large number of articles based on the TPB seem to measure the intention to act ecologically rather than behavior, primarily because validated measures of many pro-environmental behaviors do not exist (e.g., Wang et al., 2018; Whitmarsh and OöNeill, 2010). Furthermore, Stern (2000) claims that the TPB prioritizes attitude, while individual green actions are primarily influenced by knowledge and habits, which are not included in the theory. Several other factors influence pro-environmental behaviors but do not constitute an integral part of the TPB, including required effort, self-identity, connectedness to nature, and moral obligation (Gifford, 2011; Gkargkavouzi et al., 2019). In addition, factors differ from one context to another (Andersson et al., 2005), which might hinder the predictive power of the theory. For example, pro-environmental behaviors of individuals at work are affected by some factors that are absent at home—organizational values, managerial support, colleagues' attitudes, and internal culture (Yuriev et al., 2018).

In light of these limitations of TPB, a scoping review of literature was conducted to shed light on how the theory is used in these studies. Specifically, we pursued three objectives: 1) explore which individual green behaviors were studied though the lens of the TPB; 2) analyze the explained variance for intention and behavior in these studies and reveal how the TPB has been used by researchers; 3) formulate recommendations, if required, to improve the use of the theory and enhance the promotion of pro-environmental behaviors.

The next section describes the principal strengths and limitations of the TPB. Following this, methodological steps employed in this scoping review are explained. The subsequent two sections present the results of the analysis; one is dedicated to trends in the literature, and the other to the application of the theory. In the conclusion, implementation guidelines to facilitate the use of the TPB are outlined.

2. An overview of the strengths and limitations of the TPB

The success of the TPB can be explained on the basis of three strengths in its framework. First, scholars using the TPB can not only identify beliefs about the studied behaviors, but they can also consecutively assess their importance for a specific population. From a methodological point of view, direct predictors (attitude, subjective norm and PBC) are expected to be measured with a set of predefined and validated statements (Ajzen, 2002). The assessment of indirect predictors (behavioral, normative and control beliefs) requires a qualitative exploration of factors that influence a given behavior, because beliefs vary from one context to another (Ajzen, 2015). Once the beliefs associated with a given behavior are identified, researchers should conduct a set of statistical analyses with the purpose of evaluating the importance of each belief. Such analyses usually contain three elements: assessment of factors that influence behavior (in cases when the behavior is measured); evaluation of direct predictors of intention; and assessment of indirect predictors (beliefs) for significant direct predictors (Ajzen, 2002). Not only do such methodological steps allow researchers to explore similar behaviors in various contexts, but the results obtained are easy to interpret and compare. For instance, scholars working on healthcare-related issues conducted several systematic reviews and compared results of TPB-based studies (e.g., Cooke et al., 2016; McDermott et al., 2015). In the domain of pro-environmental behaviors such comparative studies do not exist; this review attempts to explore the possible reasons for this.

Second, the TPB is considered one of the most effective models for developing behavioral interventions (Riebl et al., 2015; Timm and Deal, 2016). For example, Riebl et al. (2015) analyzed the results of three TPB-based interventions that targeted unhealthy dietary habits in youth and concluded that all of them had beneficial outcomes in terms of transforming harmful behaviors. However, in the existing literature, there are few reports of interventions created using the TPB framework that aim to increase the likelihood of pro-environmental behaviors. The only example of such an intervention is reported by Timm and Deal (2016), who tested several measures for changing energy behaviors within college campuses in Illinois. Not only did the authors report a significant energy saving (up to 50%), but they also registered a non-negligible positive longitudinal effect of this intervention. This article might provide scholars with guidance on how to develop more environment-related interventions on the basis of TPB studies.

Third, the TPB is also known for its flexible structure. Although the original model is based on three direct predictors, scholars sometimes apply an extended theory by including additional variables: past behavior (e.g., Richetin et al., 2012), moral norm (e.g., Wan et al., 2017), self-identity (e.g., Mannetti et al., 2004), and multiple other variables. Some of these variables originate from theories that are frequently combined with the original TPB model: Norm Activation model (Schwartz, 1977), Value Belief Norm theory (Stern et al., 1999), the Theory of Interpersonal Behavior (Triandis, 1977), and others.

Nevertheless, scholars also point out several limitations associated with the use of the TPB. First, the theory is only suitable for exploring one behavior at a time. While it is helpful to identify specific factors that affect one action with respect to particularly desirable (e.g., physical activity, recycling, energy-saving) or harmful (e.g., use of drugs, smoking, alcohol consumption) behaviors, this approach overlooks the complexity of larger issues. For instance, corporate environmental performance depends on multiple behaviors of various employees, and one TPB-based study can hardly lead to the development of an intervention that would take into consideration the plurality of workers and their duties. According to Sniehotta et al. (2014) and Presseau et al. (2013), such situations require theoretical models that incorporate multiple goals and limited resources.

Second, the results of TPB-based studies can rarely be extrapolated because questionnaires used in such research are tailored to the target

population (Ajzen, 2002, 2011). This necessitates the development of new lists of questions, as individuals from different cultures, societies, and organizations tend to be affected by various sets of beliefs. According to certain critics of the theory (Sniehotta et al., 2005, 2014), the restricted applicability of results raises doubts about the utility of the TPB. As Sniehotta et al. (2014) state: "When have we last learned something new from reading a TPB study?" (p. 4).

Third, some scholars cast doubts over the validity of the core theory's assumptions, such as the notion that beliefs indirectly predict only intention, and not behavior. Instead, several studies have demonstrated that beliefs might also surpass intention (Araujo-Soares et al., 2013; Conner et al., 2013). Additionally, the TPB's original direct and indirect variables do not contain the multiple aspects that sometimes influence behaviors, including emotions (Rapaport and Orbell, 2000), affect (Wolff et al., 2011), regret (Conner and Armitage, 1998), and socio-economic status (Sniehotta et al., 2013).

Lastly, similar to other behavioral theories, TPB-based studies frequently employ self-reported measures of actual behavior, which might lead to biased responses and unreliable results (de Leeuw et al., 2015). In a recent review on measures of pro-environmental behaviors, Lange and Dewitte (2019) stress the importance of developing and validating assessment tools based on the contextual and psychometric properties of the studied actions. Similar recommendations are provided by Francoeur et al. (2019), who conducted a systematic literature review of items used in measuring green behaviors of employees. In fact, invalidated measures of behaviors might be one of the reasons for a so-called intention-behavior gap - the difference between respondents' intentions to act in a certain way and the actual behavior (Rhodes and Bruijn, 2013; Sniehotta et al., 2005).

Considering the growing interest in the TPB for studying pro-environmental behaviors, and considering the theory's strengths and limitations, it seemed relevant to analyze the contexts in which this theory has been used, how it has been applied, and the results it has produced. This analysis aims to outline the points that could be ameliorated in the methodological aspects of TPB's application and identify the best practices for the use of the theory.

3. Methodology

3.1. Methods

This article is based on the scoping review methodology, which is commonly used "to map the literature on a particular topic or research area and provide an opportunity to identify key concepts, gaps in the research, and types and sources of evidence to inform practice, policymaking, and research" (Daudt et al., 2013, p. 8). In contrast to systematic reviews that aim to provide answers to narrowly defined questions, the scoping approach addresses broader issues (Arksey and O'Malley, 2005). In addition, authors conducting systematic literature reviews typically assess the quality of studies prior to including them in the analyzed sample, while studies based on the scoping review methodology make conclusions about the rigorousness of included articles as a part of the analysis (Arksey and O'Malley, 2005; Daudt et al., 2013).

Although scoping reviews are used less widely than the systematic approach, several publications on topics related to sustainability demonstrate the pertinence of the scoping review approach for this study. For instance, Naylor and Appleby (2013) used a scoping review to analyze 78 articles on the implications of sustainability for the healthcare sector. Similarly, Hutchinson et al. (2015) conducted a scoping analysis of the literature the relationship between environmental and health-related behaviors.

According to Colquhoun et al. (2014), scoping reviews consist of five stages: identifying the research question(s); identifying relevant studies; selecting studies; charting the data; and collating, summarizing and reporting results. As the research questions have been outlined in the introduction, the remainder of this section discusses the other four steps.

3.2. Identifying relevant studies

In contrast to systematic literature reviews, there are no specific rules on how studies for scoping reviews should be selected. To select relevant articles for this review, a two-step process was applied. First, relevant academic journals were identified. To do so, complete lists of articles from nine systematic literature reviews on pro-environmental behaviors (Li et al., 2019; Lo et al., 2012; Koop et al., 2019; Staddon et al., 2016; Varotto and Spagnolli, 2017; Inoue and Alfaro-Barrantes, 2015; Francoeur et al., 2019; Norton et al., 2015; Yuriev et al., 2018) were extracted. These are the only existing systematic literature reviews from the last 10 years that covered various individual environmental actions. Having compiled all articles from these systematic reviews, an exhaustive list of academic journals that publish articles on the subject was created. After duplicates were eliminated, 143 journals remained (see Appendix A for the complete list).

Second, a search for pertinent articles was conducted using academic databases. The research protocol consisted of two keywords ("theory of planned behaviour" and "theory of planned behavior") and was limited to articles published before July 31, 2019. The search results yielded 905 articles.

3.3. Study selection

The studies were selected by two PhD students conducting their doctoral research projects on green behaviors. To select only those studies that employed the TPB for exploring pro-environmental individual behaviors (performed both at home and at work), the two researchers agreed upon three inclusion and exclusion criteria (see Table 1). The development of criteria is essential for literature reviews, as it allows researchers to gather only the most relevant publications (Daudt et al., 2013; Colquhoun et al., 2014). Following the criteria, first, theoretical articles were excluded, as this review targeted only empirical results. Second, included publications were expected to be focused on green actions that had a direct positive impact on the environment (recycling, energy saving, water conservation, and so on). As a result, articles on consumers (shops, products, hotels and the like) were discarded, as their activities have an indirect impact on the environment through personal preference and not actual behavior. Third, only articles in which behaviors were performed at the individual level were considered. Thus, publications that focused on initiatives adopted by organizations or departments were excluded (e.g., decision to get certified, processes optimization). This criterion also meant discarding

Table 1
Selection criteria.

Selection criteria	Inclusion criteria	Exclusion criteria
Methodology	Empirical	Theoretical
Studied behaviors	Green behaviors performed by an individual and that could have a positive impact on the environment (e.g., recycling, energy-saving activities)	Behaviors of consumers
Analysis unit	Individual	Organizational

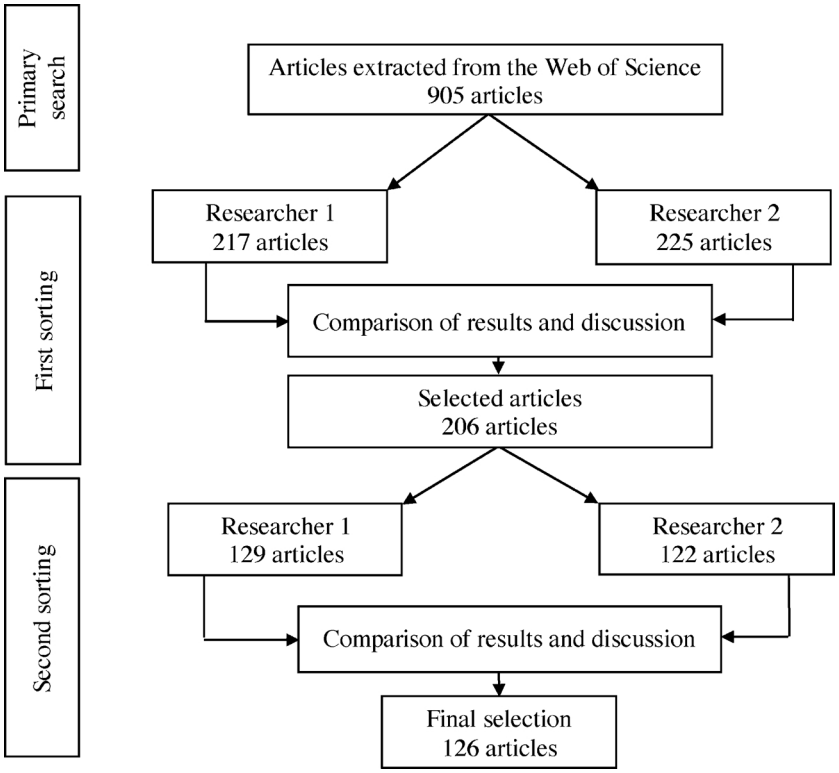


Fig. 2. Selection process.

studies about governmental institutions (e.g., new environmental policies, the effect of state incentives).

To decrease bias, the two researchers applied these criteria independently, which is considered to be the best practice for various types of literature reviews (Petticrew and Roberts, 2008; Staples and Niazi, 2007). The sorting was conducted in two steps (see Fig. 2). First, the researchers excluded articles that did not apply the TPB, but mentioned it briefly in the text. They compared their results and selected 206 articles they both agreed on. Second, researchers removed articles that did not correspond to the inclusion and exclusion criteria, which reduced the number of publications selected to 126. In both steps, the agreement level between the researchers was perfect, as independent selections were at least 95% identical (Hughes et al., 2016; Rose et al., 2012).

3.4. Charting the data

Following the recommendation of Colquhoun et al. (2014), the two researchers created a data extraction grid in Microsoft Excel. The sheet contained 18 columns that may be divided into two categories: general information about publications and data on the application of the TPB (see Table 2). Initially, the two researchers independently extracted the data from 10 articles. The results of this initial extraction were compared, and the coders discussed the few differences. To increase the reliability of the codification process, the explicit definition of each item was added to the extraction grid. Then, the researchers extracted the data from another 10 articles, and the obtained results were identical for both coders. The remaining articles were divided equally between them.

3.5. Collating, summarizing and reporting results

Similar to the data extraction, the reporting of results was divided into two steps. Primarily, the researchers analyzed literature trends by using the “general information” columns of the Microsoft Excel sheet. The following points were targeted: the popularity of the TPB over the years, the preferred study design, the diversity of studied behaviors, and the context (work/home). Then, the application of the TPB was analyzed by concentrating on the methodological steps and the reported statistical results.

Table 2
Extracted data.

General information	Application of the TPB
Article title	Dependent variables (intention and/or behavior)
Journal name	Rationale provided on why actual behavior is not studied (yes/no)
Year of publication	Direct variables (attitude, subjective norm, PBC) are measured (yes/no/partially)
Methodology	Indirect variables (beliefs) are measured (yes/no/partially)
Context (household/workplace)	Elicitation study for producing indirect variables (yes/no)
Respondent type	Names of additional variables
Studied behaviors	Rationale provided for additional variables (yes/no)
	% of explained variance attributed to the original TPB variables for intention
	% of explained variance attributed to new variables for intention
	% of explained variance attributed to the original TPB model for behavior
	% of explained variance attributed to new variables for behavior



Fig. 3. Number of articles per year and the context of studies.

4. Literature trends

4.1. Increased use of the TPB

According to the extracted data, pro-environmental behaviors have been studied through the lens of the TPB since 1995. Over the years, scholars have applied this theory more frequently; 12 articles based on Ajzen's framework were published in 2018, compared to only two in 1995 (solid line in Fig. 3). This increase might be a result of the theory's reliable predictive power in multiple situations.

4.2. Lack of research on employees' green behaviors

Although the number of articles using the TPB is growing, scholars tend to overlook the importance of the organizational context. In total, 81% of the analyzed studies were conducted within households. Only 18 publications focused on individuals' behaviors at their workplace. The gap between dashed and dotted lines in Fig. 3 reflects this difference. Despite the limited number of studies conducted within organizations, this segment of the literature is characterized by a wide variety of respondents, such as restaurant employees (Wang, 2016), construction workers (Li et al., 2018), public servants (Lam, 1999), soldiers (Laudenslager et al., 2004), and non-academic university personnel (Dixon et al., 2015; Yuriev et al., 2020).

4.3. Homogeneity of studied behaviors

Fig. 4 presents the three most frequent behavioral types that appear in the literature: recycling (29% of articles), travelling (19%), and energy saving (14%). Studies included in these categories focused on a variety of more specific behaviors. For instance, Wang et al. (2016) and Borthakur and Govind (2018) explored e-waste recycling, which was found to be influenced by slightly different variables compared with food waste recycling. Analogously, the travelling category is composed of 25 publications that explore different modes of transportation: private vehicles (Oreg and Katz-Gerro, 2006; Stanford, 2014), public transportation (Shi et al., 2017b), carpooling (Laudenslager et al., 2004), and bicycles (Muñoz et al., 2016).

Approximately one third of the analyzed publications explored other behaviors. Half of these articles (19%) studied behaviors that had not previously gained significant scholarly attention: toilet flushing (Lam, 2006), cutting trees for firewood (Serenari et al., 2011), turning off taps (Richetin et al., 2012), and drinking hot beverages (Wesselink et al., 2017). Another 19% studied pro-environmental gestures of a general nature (e.g., contribution of time and money toward ecological causes, sustainable activities, environmental activism). The imprecision in the definition of behaviors studied through the lens of the TPB is surprising, as this theoretical framework seems unsuitable for exploring non-specific actions (Ajzen, 1991, 2002). For example, Swaim et al. (2014) studied "sustainable activities," which might include numerous behaviors. Similarly, Xu et al. (2019) employed the TPB to explain "environment decision-making," which is fairly broad. In the same vein of thinking, Webb et al. (2014) explored factors that influence individuals' decision to contribute time and money toward ecological causes. While the process of donating is a rather specific behavior, multiple ecological causes cannot be accounted for within one single study. This lack of precision in the definition of studied behaviors led to ambiguity in questionnaire items and, ultimately, to misleading results.

4.4. Intention, behavior or both?

All the selected articles were based on quantitative methodologies. One third of the publications analyzed did not attempt to explain behavior and focused only on intention (e.g., Gao et al., 2017; Greaves et al., 2013). Interestingly, the titles of several papers that explored only intention give the opposite impression: "Application of the

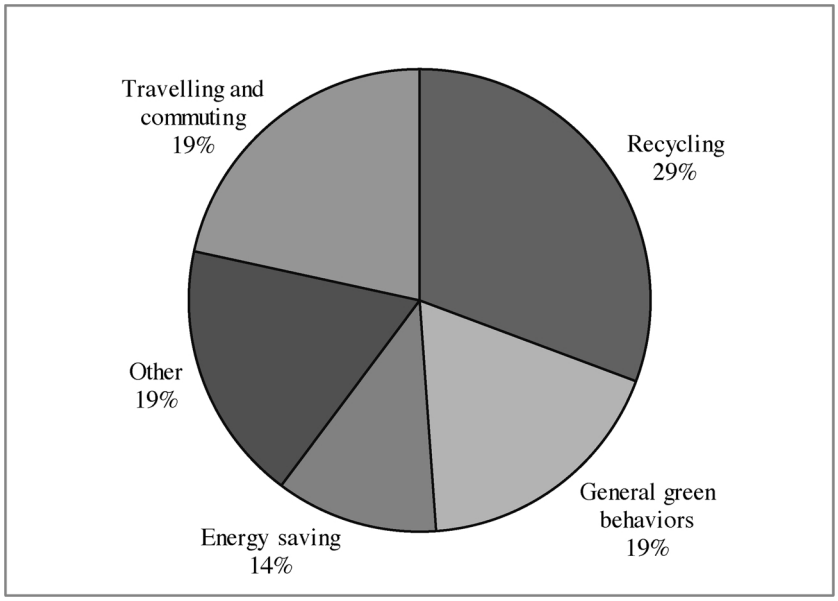


Fig. 4. Studied behaviors.

extended theory of planned behavior to understand individuals' energy-saving behavior in workplaces" (Gao et al., 2017), "Recycling: Planned and self-expressive behavior" (Mannetti et al., 2004), and "'I'm Not in the Habit of Recycling': The role of habitual behavior in the disposal of household waste" (Knussen and Yule, 2008).

Only four articles that did not study actual behaviors provide an explanation of their decision to concentrate on intentions. For instance, Wang et al. (2018) state that because "it is not practical to measure residents' actual separate collection behaviours, we researched on urban residents' intentions instead of actual behaviours to explore the dynamics of influencing factors" (p. 258). Similarly, Liu et al. (2017) claim that "it is widely recognized that behavioral intention is a key to actual behavior" (p. 43). None of the articles mentioned methodological concerns or non-existent measures of pro-environmental behavior as reasons for focusing on intention rather than behavior.

5. Application of the TPB in environmentally focused studies

5.1. The questionnaire measuring the variables of the TPB

According to the TPB, seven variables—four direct (intention, attitude, subjective norm and PBC) and three indirect (behavioral, normative and control beliefs)—have an influence on behavior. Of the analyzed articles, 90% measure direct variables, but only 14.2% measure indirect variables and 14% measure both direct and indirect variables together. Undoubtedly, the measurement of direct variables is important as it allows researchers to understand the general direction of future interventions. For example, when PBC has the largest influence, it is necessary to decrease barriers that impede the execution of the studied behavior to make individuals feel more capable of performing a particular action. Nonetheless, it is the assessment of indirect variables that is crucial for tailoring an intervention program for a given population. For instance, if PBC is the determining direct variable, scholars can measure which barriers should be prioritized, instead of addressing these barriers as a whole.

In this context, TPB-based studies should normally consist of two parts: qualitative (identification of antecedent beliefs) and quantitative (statistical assessment of direct and indirect variables). Surprisingly, out of 18 articles that measure indirect variables, only 36% describe these explorations. For example, Greaves et al. (2013) conducted several interviews with potential participants with the purpose of identifying the most pertinent beliefs that were later used in the questionnaire. Similarly, de Leeuw et al. (2015) recruited 92 individuals who wrote down their thoughts about the studied pro-environmental behaviors. Questionnaires for the other nine articles were constructed on the basis of previous studies that did not necessarily recruit similar samples of respondents. For example, Zhang et al. (2014) used five articles in developing their questionnaire. Such an approach contradicts the guidelines of Ajzen and Fishbein (1980) , according to which indirect variables have to be identified in each new context and sample. Adopting questionnaire items from previous research might lead to unreliable results.

5.2. Predictive power of the theory

Unexpectedly, although all the analyzed publications mention the objective to explain the intention, only 52% provided quantitative data about the amount of explained variance for intention. From the total of 76 studies that explored determinants of behavior, only 45% mentioned this statistic. The percentage of explained variance is the only indicator that reflects the predictive power of the TPB, and, as such, it determines whether interventions should be created on the basis of the results of the study. In this context, the omission of this information in TPB-based studies is puzzling.

The reported percentage of explained variance for intention varies between 2% and 85% (M = 44.3%, SD = 19.28), and for behavior it varies between 6% and 81% (M = 34.2%, SD = 19.23). Among the articles that indicate the percentage of explained variance for both intention and behavior (26 publications), the TPB model accounts for, on average, 24.7% more variance for intention than behavior (SD = 15.1). The minimally registered difference between the explained variance for intention and behavior is 0% (Aguilar-Luzón et al., 2012) and the maximum is 60% (Swaim et al., 2014). The analyzed articles commonly explain this gap by referring to the limitations of the methodological steps. For instance, de Leeuw et al. (2015) state that "the sample of pro-environmental behaviors used in the study was not perfectly representative of all possible PEB" (p. 137), which reduced the validity of the developed measures. In the study by Allen and Marquart-Pyatt (2018), the intention-behavior gap is believed to result from the difficulty of identifying factors that influence certain behaviors and making use of these factors in a questionnaire. The comparison of explained variance for intention and behavior in studies conducted within household and workplace contexts highlights the decreased predictive power of the TPB in the presence of multiple underlying beliefs. The average difference between explained variance for intention and behavior in articles with samples composed of employees was 27.3% (SD = 8.5), while this statistic in articles with samples composed of individuals at home was 20.2% (SD = 14.3).

5.3. Addition of new variables

The majority of the analyzed articles (72%) employ extended versions of the TPB model by introducing new variables as direct predictors of intention or behavior. In an attempt to increase the predictive power of the theory, 29% of these publications add one variable (e.g., Mannetti et al., 2004; Muñoz et al., 2016; Russell et al., 2017), while the remaining articles complement the model with up to five new predictors (e.g., Lam, 2006). The most popular additional variable is "moral norm" (27 articles), followed by "past behavior" (14 articles). "Self-identity" is introduced in ten publications, and "habit" appears seven times. Table 3 provides a list of the 10 variables that were most frequently included in the original TPB model.

6. Discussion and conclusion

6.1. General discussion

The results of this scoping review allowed us to achieve the initial objectives of the study: to understand which green behaviors are most frequently explored through the lens of the TPB; to unveil how the TPB is used in studies on pro-environmental individual actions; and to put forward recommendations to help researchers use this theory in a more systematic manner. In the first objective, the selected articles explore a wide variety of green actions. Even publications that discuss the three most frequent types of behaviors (recycling, energy saving and choosing modes of transportation) were quite diverse and concentrated on specific types of these actions, for example, e-waste recycling (Borthakur and Govind, 2018), choosing a bicycle instead of a car (Muñoz et al., 2016) and turning off computers when leaving the office (Greaves et al., 2013). The TPB domain would benefit from a profound analysis of similarities and differences between factors that influence the abovementioned types of behaviors. Considering the high volume of existing and new studies on recycling, energy saving and transportation modes, systematic literature reviews for each behavior might be necessary to fully understand underlying beliefs and their importance for various population samples. As for the context of publications, the results indicate that only a fraction of the analyzed articles (19%) study employees' pro-environmental behaviors. Hence, the variety of organizational barriers that might affect individual actions (Inoue and Alfaro-Barrantes, 2015; Norton et al., 2015; Yuriev et al., 2018) is overlooked. Nevertheless, organizations can play a significant role in promoting individual green actions; for instance, managers could

Table 3
Variables used in extended versions of the TPB.

Variable	Definition	Utility	Articles (examples)
Moral norm	"The reflection of a personal value system in a given situation" (Klöckner, 2013, p. 1030)	The notion of "right" and "wrong" is incorporated into the TPB only indirectly, through principal elements of the framework (Botetzagias et al., 2015). The idea of including moral norms is based on the fact that individuals' intentions and actual behaviors do not always depend solely on cost-benefit calculations.	Botetzagias et al., 2014, 2015; Chan and Bishop, 2013; Chu and Chiu, 2003
Past behavior	Studied behavior that the participants had performed in the past.	Individuals' decisions about behaviors are believed to frequently depend on actions performed in the past. Some past behaviors transform into habits, while others might leave an emotional trace.	Lizin et al., 2017; Mann and Abraham, 2012; Richetin et al., 2012
Self-identity	"The extent to which performing a particular role behavior is an important component of an individual's self-concept" (White and Hyde, 2012, p. 787)	Some people associate themselves with a certain identity type (for example, environmental activist), and not performing green behaviors might lead to internal dissonance.	Fielding et al., 2008; Mancha and Yoder, 2015; White and Hyde, 2012
Habit	"The automatic performance of behavioral patterns triggered by context cues—grows" (Klöckner, 2013, p. 1030)	It is considered that not all behaviors require contemplation and rational thinking. Some actions are performed in a habitual manner. Hence, "habit" is usually included in the model to predict actual behaviors, and not intentions.	Muñoz et al., 2016; Russell et al., 2017; Visschers et al., 2016
Self-efficacy	"One's perceived capability to organize and execute the course of action required to produce desired results" (Bandura, 1997, p. 2)	In comparison with PBC that emphasizes the level of control ("If I want, I can"), self-efficacy refers to the feeling of confidence that a person possesses in respect to a given behavior ("I believe in myself"). Due to the complementary meanings of PBC and self-efficacy, scholars frequently include self-efficacy items in the measurement of PBC.	Shi et al., 2017a, b; Tang et al., 2011
Anticipated emotions	Various emotional states that follow the performance of a certain behavior.	Emotional experiences (regret, happiness, joy, and so on) might have a positive or negative effect on people's willingness to repeat behaviors. Even when individuals perform a behavior for the first time, they might be affected by perceived emotions associated with the contemplated action.	Russell et al., 2017; Webb et al., 2013, 2014
Environmental awareness	"Environmental knowledge and the recognition of environmental problems" (Blok et al., 2015, p. 57)	When individuals are aware of environmental problems, they are presumably more prone to acting in a responsible manner. Traditional constructs of the TPB only include this knowledge indirectly.	Blok et al., 2015; Echegaray and Hansstein, 2017; Wan et al., 2017
Environmental values	"An ordered set of beliefs about desirable end states that guide the selection or evaluation of environmentally relevant behaviors" (Morgan et al., 2015, p. 36)	While attitudes are situation-specific, values tend to stay unchanged for a long period of time. According to Thompson and Barton (1994), there are three types of such values: ecocentrism, anthropocentrism, and environmental apathy.	Blok et al., 2015; Kaiser et al., 1999; Oreg and Katz-Gerro, 2006
Sense of community	The feeling of an affiliation with a collective within the organization (Dixon et al., 2015)	Subjective norms are considered to have stronger influence when individual's sense of community is higher, as opinions of others will be more important.	Dixon et al., 2015; Tang et al., 2011
Socio-economic and demographic factors	Variables such as gender, age, income, education, region of residence, and so on.	These variables are included to analyze participants' intentions and actual behaviors depending on their socio-economic or demographic status. Although they are not directly inserted into the TPB model, they might help scholars to get a deeper understanding of factors that predict intentions and behaviors.	Blok et al., 2015; Echegaray and Hansstein, 2017; Zwickle et al., 2016

encourage employees to act responsibly by being role models (Boiral et al., 2015), workers could be asked to set their own environmental goals (Tosti-Kharas et al., 2016; Yuriev et al., 2020), and rewards and recognition schemes for ecological gestures could be launched (Lamm et al., 2013; Manika et al., 2015; Ramus, 2002). In this sense, identifying and assessing specific barriers associated with the workplace context could be crucial for achieving better employee engagement and a lower ecological footprint for organizations. The current lack of TPB-based research in organizational contexts leads to a poor understanding of specific factors that influence employees' pro-environmental behaviors.

Our second goal was to understand how scholars used various variables of the TPB in studies on pro-environmental behaviors and to analyze how well the theory predicts intention and behavior. This analysis points out two striking issues associated with the use of the theory. First, the majority of publications (86% of the selected articles) overlook indirect variables of the theory (behavioral, normative and control beliefs). Without exploring specific factors that influence behaviors, these studies undermine the principal utility of the TPB—identifying beliefs that should be targeted in behavioral interventions. Furthermore, only six articles describe elicitation studies that are supposed to be conducted prior to the quantitative phase in order to identify underlying beliefs and develop a questionnaire. In this context, the measures of indirect predictors adopted in certain publications do not seem reliable, as each population is affected by its own set of beliefs (Ajzen, 2002). Moreover, studies that overlook the importance of indirect variables could only identify the direct variables that should be addressed, and thus they are less effective for the development of tailored intervention plans. Second, despite the quantitative core of any TPB-based study, only half of the analyzed publications provided results of the explained variance for intention or behavior. The explained variance is the principal indicator of a study's ability to detect the most influential variables of the studied phenomena. Given the importance of the explained variance, its omission in over 40 articles is puzzling. Nevertheless, reported numbers in publications that contain such statistics are similar to TPB-based research from other domains. Our analysis indicated that the average explained variance for intention was 45.9% and the average explained variance for behavior was 37.2%. In comparison, Godin and Kok (1996) analyzed 56 articles on health-related behaviors and reported the

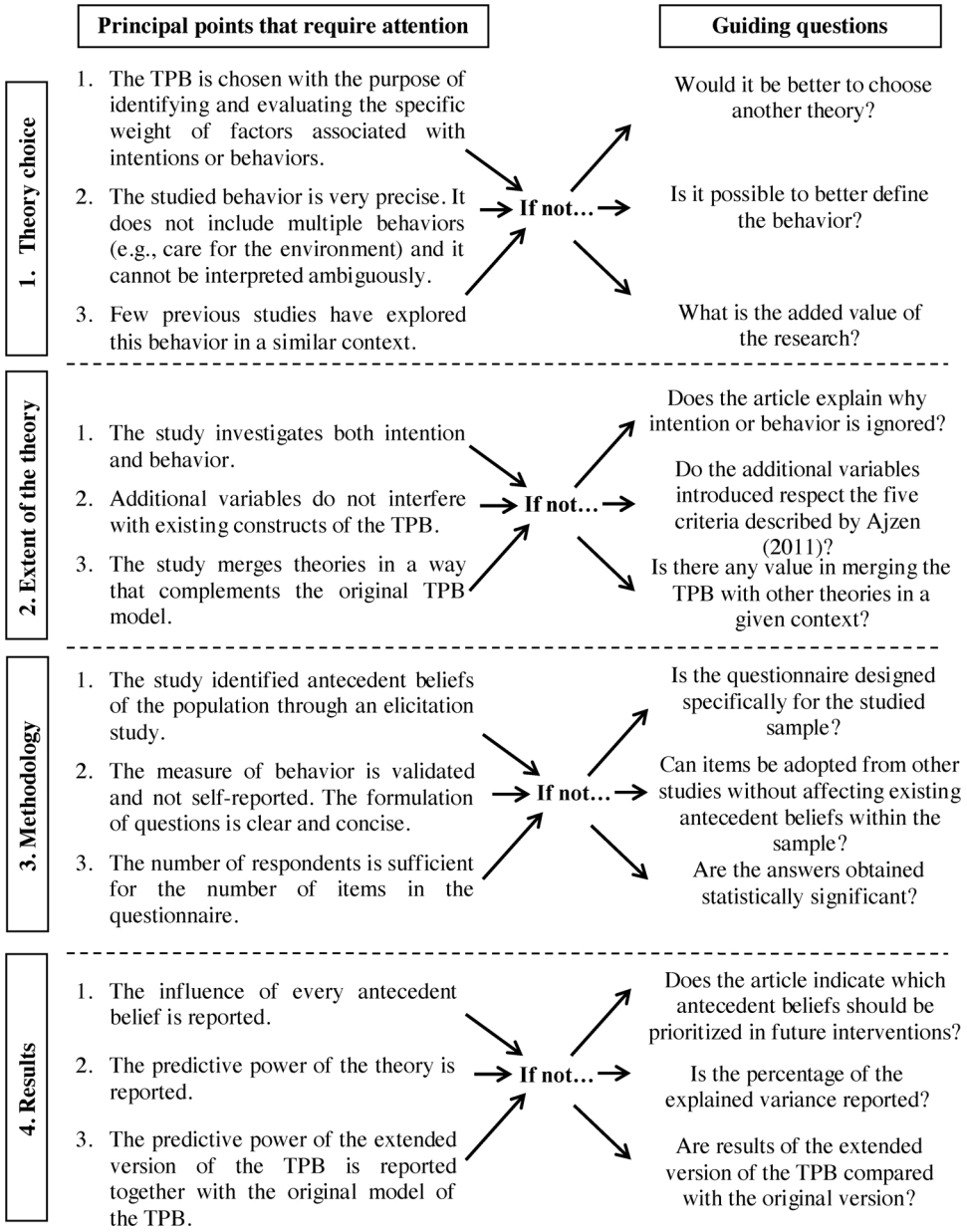


Fig. 5. Roadmap for avoiding common pitfalls when implementing the TPB.

average explained variance as 41% for intention and 34% for behavior. It is worth noting that the review by Godin and Kok (1996) was conducted over 20 years ago, and since then researchers have suggested various new variables to increase the predictive power of the theory. The analysis of selected publications indicates that studies with additional variables always perform better and increase, on average, the explained variance for intention by 12.1% and for behavior by 10.5%. Although the authors of the analyzed articles most frequently complemented the original TPB model with "moral norm," "past behavior," and "self-identity," some additional variables are specific to pro-environmental behaviors (for example, environmental values and environmental awareness). Such domain-specific predictors should be prioritized in studies on green behaviors because more studies are required to draw convincing conclusions on the increased predictive power of extended versions of the TPB for pro-environmental behaviors.

Our third goal was to formulate recommendations that would allow researchers to ameliorate the use of the TPB. Generally speaking, the abovementioned issues with the adoption of the theory can be divided into four categories: choice of framework, decision to extend the TPB model, methodology, and results. To help researchers overcome these issues, a roadmap was developed (Fig. 5). The objective of this tool is not to act as a substitute for existing texts about the theory's application, but to help researchers working on pro-environmental behaviors avoid the common issues identified in this literature review. Scholars working on green behaviors are encouraged to follow this roadmap when applying the TPB, as it will allow them to question several crucial aspects that seem to be frequently overlooked in the domain of pro-environmental actions.

6.2. Limitations and avenues for future research

The principal limitation of this scoping review lies in the selection of articles. The list of reviewed journals was composed on the basis of nine literature reviews on pro-environmental behaviors. These literature reviews had their own inclusion and exclusion criteria, and this may have reduced the list of journals ultimately selected. However, considering that 143 journals were screened, this limitation does not seem

to skew the results. On the contrary, the analysis of the data offers two promising avenues for future research.

First, future studies should put more effort into overcoming the intention-behavior gap when it comes to green behaviors. Currently, scholars seem to focus on explaining factors that influence intention and not those that influence behavior. The results of the present scoping review indicate that the average difference between the explained variance of intentions and behaviors in articles on pro-environmental behaviors is 22.5%. Nevertheless, this should not be perceived as a limitation, but as an interesting result that requires interpretation. While some studies indicate that this might be a result of habit (Knussen and Yule, 2008) or forgetfulness (Ajzen, 2012), it is necessary that researchers concentrate on closing the gap. Initial efforts should be devoted to developing validated measures for measuring pro-environmental behaviors (Wang et al., 2018; Whitmarsh and O'Neill, 2010), which would allow researchers to identify and evaluate factors that influence actual behaviors with more precision. In general, it might be fruitful to investigate the intention-behavior gap through longitudinal studies that are currently lacking in the domain of green behaviors. Alternatively, this issue might be addressed by integrating the foundations of the TPB into a different conceptual framework. Articles where the TPB is combined with other theories reported higher explained variance (Bamberg et al., 2003, 2007); hence, researchers can attempt to insert multiple variables from Table 3, simultaneously, into their studies.

Second, scholars should not underestimate the role of organizations in the promotion of sustainable actions, as organizational obstacles might be as significant as individual factors (Greaves et al., 2013; Norton et al., 2012; Yuriev et al., 2018). In this context, a question that begs additional research is "What organizational barriers should be prioritized and why?" The current literature on obstacles to pro-environmental behaviors simply identifies the existence of many such factors, whereas it is necessary to have a thorough understanding of specific obstacles that constitute a priority for organizations of different sizes. The TPB can be helpful for achieving this goal as it allows scholars to quantitatively evaluate a plethora of barriers. However, the results of the present scoping review indicate that such factors are rarely assessed, thus undermining one of the principal strengths of this theoretical model. The lack of TPB-based studies on pro-environmental behaviors in the organizational context can also be used to inform policymakers. Specifically, organizations can play a significant role in the promotion of green actions among their employees (e.g., Boiral et al., 2015; Smith and O'Sullivan, 2012). Recent research demonstrates that employees who behave in an eco-friendly way at work are more prone to being environmentally conscious at home (Paillé et al., 2019). In this context, policymakers should encourage scholars to conduct more studies on pro-environmental behaviors of employees by providing financial support and facilitating access to data.

In sum, this study demonstrates the importance of following the guidelines for using the TPB model. Researchers should clearly identify the reasons why the theory was selected and specify the extent of its application in their articles. Further, it is not sufficient to suggest a new variable because it has never been introduced before. Without respecting the set of criteria outlined by Ajzen (2011), studies on pro-environmental behaviors will stagnate. The TPB requires advanced methodology that includes both qualitative and quantitative elements; skipping any of these methodological steps could render results unreliable because they would not be specific to the studied population. In addition, results should be reported in a consistent manner: information about explained variance should be indicated clearly and completely. To this end, the roadmap proposed in this article (Fig. 5) might help scholars achieve more consistent results when using the TPB.

Declaration of interest

None.

Appendix A. List of selected academic journals

1. Academy of Management Journal
2. Accounting, Auditing & Accountability Journal
3. American Economic Review
4. American Journal of Community Psychology
5. American Psychologist
6. Applied Energy
7. Applied Environmental Education & Communication
8. Applied Psychology: An International Review
9. Architectural Engineering and Design Management
10. Architectural Science Review
11. Automation in Construction
12. Basic and Applied Social Psychology
13. Behaviour & Information Technology
14. British Journal of Mathematical and Statistical Psychology
15. British Medical Journal
16. Building Research & Information
17. Building Simulation
18. Business Process Management Journal
19. Business Strategy and the Environment
20. California Management Review
21. Canadian Psychology/Psychologie Canadienne
37. Environmental education research
38. Environmental science & policy
39. European Management Journal
40. European Planning Studies
41. European journal of social psychology
42. Evaluation Review
43. Facilities
44. Frontiers in Psychology
45. Geographical Research
46. Global environmental change
47. Group & Organization Management
48. Habitat International
49. Health Information & Libraries Journal
50. Health Psychology Review
51. HortTechnology
52. Human Organization
53. Human ecology
54. Implementation science
55. India Review
56. Information Systems Journal
57. Intelligent Buildings International
73. Journal of Consumer Affairs
74. Journal of Consumer Policy
75. Journal of Economic Psychology
76. Journal of Economic Surveys
77. Journal of Environmental Management
78. Journal of Environmental Psychology
79. Journal of Environmental Systems
80. Journal of Experimental Social Psychology
81. Journal of Industrial Ecology
82. Journal of Management
83. Journal of Marketing
84. Journal of Marketing Communications
85. Journal of Marketing Management
86. Journal of Occupational Health Psychology
87. Journal of Organizational Behavior
88. Journal of Solid Waste Technology and Management
89. Journal of Supply Chain Management
90. Journal of Sustainable Tourism
91. Journal of Transport Geography
92. Journal of Urban Planning and Development
93. Journal of World Business
109. Organization & Environment
110. PLOS One
111. Perceptual and Motor Skills
112. Population and Environment
113. Pratiques Psychologiques
114. Psychological Reports
115. Psychological bulletin
116. Psychology & Marketing
117. Public Performance & Management Review
118. Public health
119. Renewable and Sustainable Energy Reviews
120. Resources, Conservation and Recycling
121. Review of Economics and Statistics
122. Social Indicators Research
123. Social influence
124. Society and natural resources
125. Sustainable Development
126. Sustainable Production and Consumption
127. Technology Analysis & Strategic Management
128. The Academy of Management Journal
129. The American Review of Public Administration

22. Cellulose Chemistry and Technology	58. International Journal of Consumer Studies	94. Journal of Consumer Culture	130. The International Journal of Human Resource Management
23. Climatic Change	59. International Journal of Environmental Research	95. Journal of the American Statistical Association	131. The International Journal of Sustainable Development & World Ecology
24. Conservation Letters	60. International Journal of Environmental Research and Public Health	96. Journal of the American Water Resources Association	132. The Journal of Environmental Education
25. Construction Management and Economics	61. International Journal of Hospitality Management	97. Journal of the Association of Environmental and Resource Economists	133. The Journal of Socio-Economics
26. Current Opinion in Psychology	62. International Journal of Sustainability in Higher Education	98. Landscape and Urban Planning	134. The World Bank Economic Review
27. Ecological Economics	63. Journal of Advertising	99. Local Environment	135. Tourism Management
28. Educational psychologist	64. Journal of Agricultural and Resource Economics	100. Long Range Planning	136. Transportation Research Part A: Policy and Practice
29. Energy Efficiency	65. Journal of Ambient Intelligence and Humanized Computing	101. Management Decision	137. Utilities Policy
30. Energy Policy	66. Journal of Applied Behavior Analysis	102. Manufacturing & Service Operations Management	138. Waste Management & Research
31. Energy Research & Social Science	67. Journal of Applied Psychology	103. Media Psychology	139. Water
32. Environment and Behavior	68. Journal of Applied Social Psychology	104. Mediterranean Journal of Social Sciences	140. Water International
33. Environment and planning	69. Journal of Business Ethics	105. Motivation and Emotion	141. Water Policy
34. Environment, development and sustainability	70. Journal of Business Research	106. NJAS-Wageningen Journal of Life Sciences	142. Water Resources Research
35. Environmental Communication	71. Journal of Cleaner Production	107. Natural hazards	143. Water Science and Technology
36. Environmental Modelling & Software	72. Journal of Computing in Civil Engineering	108. Nature	

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